Xiaochuan Li

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RESEARCH FOCUS

I study how people learn from linguistic experience and how that experience shapes comprehension and reasoning. My master's thesis is a behavioural study with preschool children; alongside it I use computational models to quantify regularities in linguistic input and test hypotheses about human cognition.

EDUCATION

University of Technology Nuremberg — **Nuremberg, Germany**
M.Sc. Human and Artificial Intelligence — Oct. 2025 – exp. 2027
- Coursework: psychology and empirical study design, cognitive neuroscience, deep learning, language models

University of Marburg — **Marburg, Germany**
German for graduate study; DSH-2 (C1 level) — Oct. 2024 – Jul. 2025

Shenzhen University — **Shenzhen, China**
B.A. Philosophy — Sep. 2020 – Jun. 2024
- Coursework: formal and modal logic, the formal background to my M.Sc. thesis on modal language; philosophy of language
- B.A. thesis (May 2024): Leibniz's logical system and *characteristica universalis*

RESEARCH EXPERIENCE

Modal language development in preschoolers — Sep. 2026 – Present
M.Sc. thesis, Max Planck Institute for Human Cognitive and Brain Sciences; supervisor: Charlotte G.
- How preschoolers come to distinguish possibility from necessity in language, and how this relates to their nonlinguistic reasoning about possibilities. Behavioural study, in design

The direction of disagreement: what departures from the majority term teach a text-only learner — Jul. 2026 – Nov. 2026
Independent project
- Asks what a learner can infer from speakers' disagreement about where a word's boundary lies. Learners read only naming sentences, with the disagreement kept or randomly replaced, and are tested on whether they recover the order of categories
- Methods: formal proof; transformers and LSTMs trained from scratch on controlled input; World Color Survey
- Preprint: lingbuzz/010504 · doi:10.5281/zenodo.23677586

PROJECTS

Immediate repetition and familiarity-like EEG representations of unfamiliar faces — Oct. 2026 – Present
Group study in the Cognitive Neuroscience course, reanalysis of open data
- Preregistered representational similarity analysis of open EEG data (16 participants, 70 channels)

How the first information source shapes verification: a between-subjects experiment — Dec. 2025
Class experiment in the course Psychology and Empirical Study Design
- Between-subjects experiment on Pavlovia (49 participants, 47 analysed). My part: experimental design, in a group of three, and the data analysis in JASP

Moral framing and dilemma choices in language models — Apr. 2026 – Nov. 2026
Group project supervised at the German Research Center for Artificial Intelligence (DFKI)
- Designed minimal pairs to measure how moral framing shifts model choices; ran the utilitarian-framing branch, including activation patching and probing

EMPLOYMENT

University of Technology Nuremberg — **Nuremberg, Germany**
IT Service Desk (part-time) — Apr. 2026 – Present
- Student IT support; basic HPC user administration (resource allocation, access rights)

Shenzhen Learning Disorders Association — **Shenzhen, China**
Internship, learning assessment at Yucai No. 2 Primary School — Sep. 2021 – Dec. 2021
- Prepared character-reading materials, administered timed reading assessments to primary-school children referred for learning difficulties, and analysed the response data; contributed to individual learning plans

SKILLS AND LANGUAGES

- **Experimental design and analysis:** between-subjects designs, PsychoPy/Pavlovia, JASP
- **Computational modelling:** training models from scratch on controlled input; minimal-pair evaluation; activation patching, probing
- **Programming:** Python (pandas, scikit-learn, PyTorch), Git, LaTeX, Slurm on an HPC GPU cluster
- **Languages:** Mandarin Chinese (native), German (C1), English (C1), Japanese (learning)